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DEPARTMENT OF DATA SCIENCE

IV CSD-C (Data Science) Major Project – Design Phase Progress Report

A.Y. 2026-2027

1. Project Details

Team No	A20	**Review / Report Date**	12-10-2026
Title of the Project	CampusIQ: An AI-Powered University Student Support and Academic Intelligence Platform		
Domain / Areas	Artificial Intelligence	**DS Task Type**	*NLP / GenAI*
Guide Name	Mrs. Arshiya Begum	**Reporting Period**	From: _____ To: _____
SDG Goal(s) Mapped	SDG 4 – Quality Education; SDG 10 – Reduced Inequalities		

S.No	Roll No	Name	Role / Module Owned
1	23R21A6754	Shaik Abdul Kareem Ayemen	Module 1 — Intelligent Knowledge Base & RAG Engine
2	23R21A6755	Sai Krishna	Module 2 — Personalized Student Intelligence
3	23R21A6704	Shaik Sameer Basha	Module 3 — Student Portal, Administration & Integration

2. Design Activity Status

**Status: C = Completed | IP = In Progress | NS = Not Started. Ref C1–C8 maps to the Design Phase Review Rubric.*

Ref	Design Activity	Deliverable / Evidence	Status*	% Done	Remarks
C1	**Problem formulation & requirements**	SRS: DS task, target variable, success metrics, FR / NFR	C	100	Defined the need for centralized and personalized university support.
C2	**Literature review & baseline**	Survey table (min. 10 papers), baseline results, research gap	C	100	Studied RAG-based academic assistants and university chatbots.
C3	**Data acquisition & understanding**	Dataset finalised, data dictionary, EDA report	C	100	Identify and collect authorized university documents and permitted student data.
C4	**Preprocessing & feature engineering design**	Preprocessing pipeline & feature plan	C	100	Define extraction, cleaning, chunking, metadata and embedding steps.

Ref	Design Activity	Deliverable / Evidence	Status*	% Done	Remarks
C5	**Model / algorithm design**	Candidate model comparison & selection rationale	C	100	Compare keyword search, semantic retrieval, RAG and rule-based eligibility analysis.
C6	**System architecture & ML pipeline**	Architecture diagram, DFD / UML, tech stack	C	100	Divide the platform into three modules with defined API boundaries.
C7	**Evaluation & validation strategy**	Metrics, data split / CV plan, test cases	IP	56	Plan retrieval, response, eligibility and end-to-end tests.
C8	**SDG alignment & responsible AI**	SDG mapping, bias / privacy / explainability checklist	IP	34	Plan access controls, source citations and privacy safeguards.
—	**Environment & tools setup**	Repository (GitHub), compute (Colab / GPU), libraries	IP	40	Confirm repository, dependencies and development setup.
	Overall Design Phase Completion			73%	

3. Summary of Work Done in Design Phase

Briefly describe the refined problem statement, key design decisions taken and how the design meets the requirements (150–250 words).

CampusIQ is an AI-powered university student support platform designed to simplify access to academic and administrative information. It uses RAG, NLP, LLMs, and semantic search to provide accurate, context-aware responses from university documents. The project is divided into three modules: Intelligent Knowledge Base & RAG Engine, Personalized Student Intelligence, and Student Portal, Administration & Integration.

Module 1 handles document processing, information retrieval, and AI-generated responses with source references.

Module 2 provides student-specific guidance, eligibility analysis, personalized recommendations, and notifications.

Module 3 develops the user interface, authentication, dashboards, grievance management, and integration of all modules.

The proposed technology stack includes React.js, TypeScript, Node.js, Express.js, Python, FastAPI, and MongoDB. Overall, CampusIQ aims to improve information accessibility, personalize student support, and make university services more efficient.

4. Dataset Details

Dataset Name	Academic AI Assistant — Knowledge Retrieval Challenge
Source / URL	https://www.kaggle.com/competitions/machine-deep-learning/data
Size (records × features / images / text corpus)	190 knowledge-base records across 10 JSON files, with 21 distinct field names across the different record types. The dataset also contains 110 labelled training queries, 50 unlabelled test queries, and 50 rows in the sample-submission file. The ZIP is 8,144 bytes (about 7.95 KiB); its extracted contents total 26,653 bytes (about 26.03 KiB).
Data Type (tabular / image / text / audio / time-series)	Structured JSON and CSV text data. Includes faculty details, courses, examination schedules, academic calendar events, attendance and placement policies, campus services, holidays, hostel rules, FAQs, and natural-language queries.
Licence / Permission	Subject to Competition Rules. Kaggle does not list this dataset as having an unrestricted open-data licence. The competition rules specify eligibility and usage restrictions, so permission for reuse in a separate academic project should be confirmed before adopting it.
Target Variable / Labels	The document_id field in train_queries.csv identifies the correct knowledge document for a given query. This label can be used to evaluate document retrieval.
Known Issues (missing values, imbalance, noise)	The knowledge base represents a fictional academic institution and may not reflect the actual rules and policies of the project team's university. The dataset also does not provide real student academic records for personalized eligibility analysis.
Train / Validation / Test Split	Use the provided labelled training queries for retrieval development and validation, and the unlabelled test queries for retrieval evaluation after obtaining verified ground truth. Create a separate validation split from the labelled training data to avoid using the test set during development.

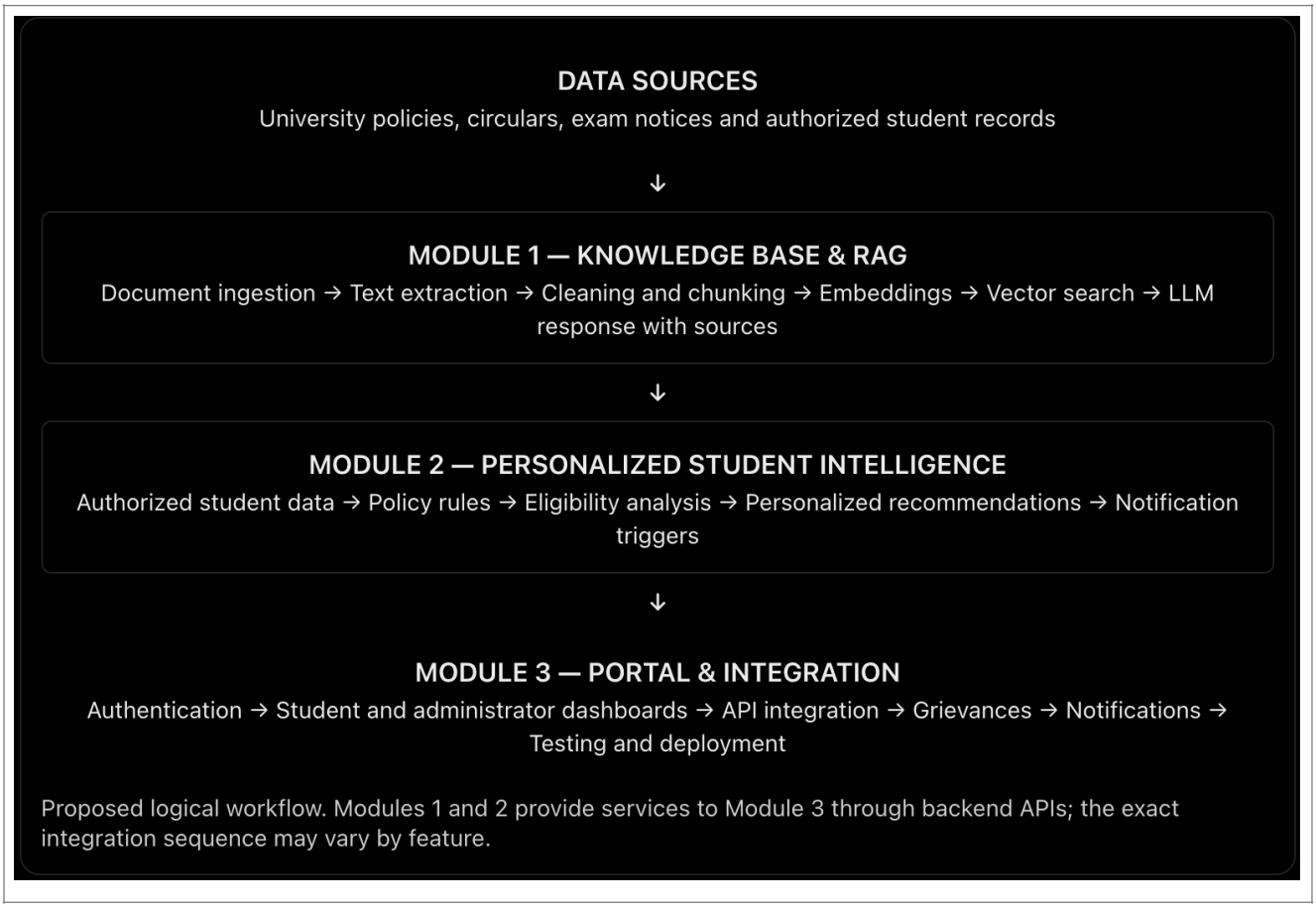
5. Proposed Models / Approach

S.No	Model / Algorithm	Rationale for Selection	Baseline / Proposed	Evaluation Metric(s)
1	Keyword-based document retrieval	Establishes a simple baseline for matching student queries with documents.	Baseline — Module 1	Precision, Recall
2	Embedding-based semantic search	Retrieves relevant passages even when query wording differs from document wording.	Proposed — Module 1	Precision, Recall, MRR

S.No	Model / Algorithm	Rationale for Selection	Baseline / Proposed	Evaluation Metric(s)
3	Retrieval-Augmented Generation (RAG) with an LLM	Generates context-aware responses grounded in retrieved university sources.	Proposed — Module 1	Answer correctness, faithfulness, citation accuracy
4	Rule-based eligibility engine	Applies verified institutional criteria to authorized student data for transparent eligibility decisions.	Proposed — Module 2	Accuracy, precision and recall against verified test cases

6. System Architecture / ML Pipeline

Insert the end-to-end architecture diagram (data ingestion → preprocessing → training → evaluation → deployment / dashboard).



Tech Stack:

- Language - TypeScript, JavaScript, Python
- Libraries / Frameworks - LangChain or an equivalent framework
- Database / Storage - MongoDB
- Deployment - Vercel, Render

7. Individual Contribution

Roll No	Name	Design Tasks Completed	Hours	Signature
23R21A6754	Shaik Abdul Kareem Ayemen	Module 1: RAG architecture, literature review related to retrieval, document-processing plan, embedding and vector-search design, response evaluation strategy.	30	
23R21A6755	Shaik Sameer Basha	Module 2: Student-data schema, eligibility rules, personalization workflow, notification logic, privacy requirements and eligibility test-case design.	27	
23R21A6704	Sai Krishna	Module 3: UI and dashboard planning, authentication and role design, backend API contracts, grievance workflow and integration architecture.	25	

8. Challenges Faced & Mitigation

S.No	Challenge / Risk	Impact	Mitigation / Support Needed
1	Extracting and organizing information from university documents with different formats	Poor text quality may reduce retrieval relevance and answer accuracy.	Use preprocessing, document chunking, metadata tagging, source validation and document-version tracking.
2	Integrating the RAG engine with student-specific eligibility services	Inconsistent data formats or API failures may produce incomplete or incorrect personalized responses.	Define API contracts, validate data, use verified policy rules and perform integration testing with controlled test records.
3	Protecting student data and preventing unsupported AI responses	Privacy violations or misleading guidance could reduce reliability and user trust.	Apply role-based access, least-privilege permissions, data minimization, source citations, validation checks and fallback responses.

9. Plan for Next Phase (Implementation)

S.No	Implementation Task	Owner	Start Date	End Date
1	Implement document ingestion, preprocessing, embeddings, semantic search and the RAG API.	Shaik Abdul Kareem Ayemen	13-10-2026	18-10-2026
2	Implement authorized student-data integration, eligibility rules and personalized recommendations.	Sai Krishna	13-10-2026	18-10-2026
3	Develop student and administrator interfaces, authentication and role-based dashboards.	Shaik Sameer Basha	13-10-2026	18-10-2026
4	Integrate the RAG and student-intelligence APIs with the portal; connect notifications and grievance workflows.	All three members	19-10-2026	25-10-2026
5	Conduct functional, retrieval, eligibility, security and usability testing; fix issues and prepare the final demonstration.	All three members	19-10-2026	25-10-2026

10. Guide Assessment

Guide's Comments	
Progress Rating	☐ Excellent ☐ Satisfactory ☐ Needs Improvement ☐ Unsatisfactory
Ready for Implementation Phase?	☐ Yes ☐ Yes, with modifications ☐ No – re-review required

_____	_____	_____	_____
Team Leader	**Guide**	**Project Coordinator**	**HOD (CSE-DS)**